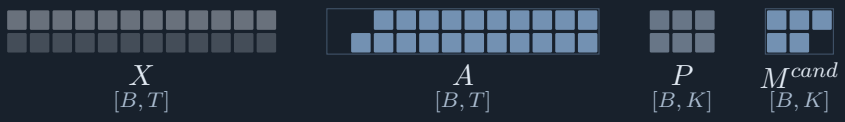
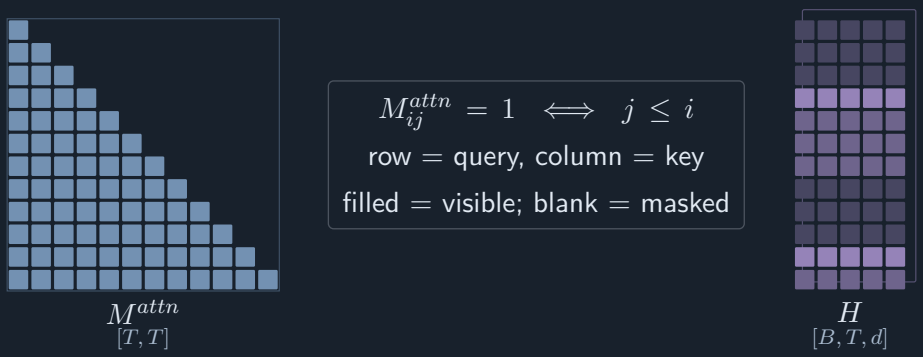
$$X \in \mathbb{Z}^{B\times T}, \quad H \in \mathbb{R}^{B\times T\times d}, \quad S \in \mathbb{R}^{B\times K}, \quad \tau > 0$$

Discrete inputs: token IDs, padding support and candidate indexing



Attention support is distinct from padding and candidate support



Tensor geometry follows the final two axes; transposition exchanges their edges



Axes B: questions; T: tokens; d: hidden channels; V: vocabulary; K: candidate columns; L: answer IDs; $R = \sum_b K_b$: pair rows. C: packed sequences; Q: questions per packed sequence.

Meaning A matrix face shows its last two axes; offset sheets indicate leading batch axes. Vectors are drawn as columns. Binary support diagrams show exact illustrative support. τ is temperature; T remains sequence length.

Cell counts illustrate axes, not model sizes; tint identifies roles, not numeric values.